Three Ways an Elicited Evaluation Measures Its Own Design

Item sampling, instruction effects, and threshold artifacts,
with a worked case and 7,920 elicitations

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Abstract

A large class of evaluations asks a language model to report a number about a statement—a confidence, a rating, a degree—and treats that number as a measurement. We quantify three ways such a design measures itself rather than the model, on 7,920 elicitations across six frontier model families and two unrelated construct families.

First, item sampling. A rate computed on one statement is a property of that statement: between-item standard deviations exceed the mean in four of five constructs, so a single-item probe carries a 95% interval half-width of ± 0.41 on a $[0, 1]$ rate. A published figure of 0.661 from one sentence becomes 0.222 over ten; reaching ± 0.05 takes 69 items in one family and 20 in the other.

Second, instruction effects, where the obvious ablation misleads. Deleting the sentence that names the measured behaviour barely moves the result; deleting the surrounding framing removes it. Running only the first half would have exonerated the instrument.

Third, threshold artifacts. Models answer on a coarse grid, so a cut on a modal value adjudicates by rounding: moving one cut from 0.60 to 0.90 changes a rate by a factor of 16 without touching the data.

Replicating all three on an unrelated task shows that no magnitude transfers intact and that the framing effect does not replicate in the mean, though it still moves the response shape, which sharpens the rule. We state the design rule each implies and release both banks, the code and every generation.

1 The shape of the problem

Consider an evaluation of the following kind. You have a construct—hallucination, refusal, value conflict, ambiguity, harmfulness—and you want to know how often a model exhibits some response to it. You write one or a few statements that instantiate the construct, you ask the model a question about each, you record a number, and you report the rate. Variants of this design are everywhere: verbalized confidence [Lin et al., 2022], self-reported uncertainty, LLM-as-judge scoring [Zheng et al., 2023], self-critique, constitutional-style self-evaluation, and most bespoke probes built for a paper.

That the resulting numbers are treated as measurements without the apparatus measurement usually requires is a concern raised both inside NLP [Bowman and Dahl, 2021] and from measurement theory [Jacobs and Wallach, 2021]. The design has three failure modes that are individually well known and jointly under-measured. The items may not represent the construct. The instruction may produce the behaviour it measures. And the decision rule may sit where the model's answers pile up. This paper measures all three on one corpus, so that the three magnitudes can be compared against each other rather than asserted in isolation, and then measures them again on a second corpus that shares the design and nothing else, so that the three can be told apart from the subject matter.

The first corpus comes from a study of a four-valued logic, reported elsewhere [Smarandache and Leyva-Vázquez, 2026b,a]. That study's subject matter is not the subject matter here; we use it because it happens to have the design features needed—a purpose-built item bank with ten matched items per construct, six model families, an instruction that can be ablated in two nested steps, and a decision rule that is a threshold on a sum. Readers who care about the logic should read those papers. Readers who run elicited evaluations should find everything they need here.

The second corpus was built for this paper and shares none of that content. It asks the most ordinary elicited question there is—how confident are you that this statement is true—over well-known and obscure facts, well-known and obscure falsehoods, and genuinely open questions, with the same models, the same wordings and the same two nested ablations. It exists because a methods paper resting on one subject matter cannot distinguish a property of the design from a property of that subject matter, and Section 7 shows the distinction is not academic: one of the three effects does not survive the move.

Relation to the companion papers. This is a methods paper and it makes no claim about the logic under test there. Its contributions are the three quantities below, the design rules that follow from them, the ablation protocol in Section 4, and the whole of the second construct family, which is new data collected for this paper and appears in neither companion. Where a number is reported in a companion paper we say so.

What is not new. That elicitation format moves elicited confidence is established [Kim and Kang, 2026], as is the unreliability of LLM judges [Yagubyan, 2026]; that prompt formatting alone moves accuracy by many points [Sclar et al., 2024]; that few-item instruments are imprecise is elementary psychometrics [Cronbach et al., 1972]; and that evaluation results deserve intervals is argued directly for language models by Miller [2024]. What we add is the size of each effect measured on a common corpus, the fact that the second one defeats the natural ablation, the conversion of the first into a number of items, and a replication on an unrelated construct family that shows which of the three magnitudes transfer—none of them—and why.

2 Corpus

All results below come from elicitations against six frontier model families, one per vendor, through a single router at temperature 1.0: GPT-4o, Claude Sonnet 4, Llama 4 Maverick, DeepSeek Chat, Qwen3 235B and Mistral Medium 3.1.

In Family I the outcome we track is a binary label derived from the reported numbers by a threshold rule. Its content does not matter here; what matters is that it is a rate computed over items, which is the form almost every elicited evaluation takes. In Family II we report the reported number itself, for a reason that is one of the findings: on that bank the threshold rule destroys the measurement rather than summarising it (Section 7).

Study	Design	Elicitations
Family I — epistemic states, four components		
Bank	110 items $\times$ 6 models $\times$ 3 wordings	1,980
Triple	110 items $\times$ 6 models $\times$ 3 reps	1,980
Pilot	8 items $\times$ 6 models $\times$ 3 wordings $\times$ 10 reps	1,440
Ablation A	60 items $\times$ 6 models, instruction reduced	360
Ablation B	60 items $\times$ 6 models, framing removed	360
Family II — factual confidence, one component		
Factual bank	60 items $\times$ 6 models $\times$ 3 wordings	1,080
Ablation A$'$	60 items $\times$ 6 models, instruction reduced	360
Ablation B$'$	60 items $\times$ 6 models, framing removed	360
		7,920

Table 1: The corpus. Family I holds ten matched statements for each of five constructs—ethical conflict, epistemic ignorance, vagueness, future contingency and logical paradox—plus ten anchors of settled truth. Family II (Section 7) is an unrelated task in the same elicited form: ten statements for each of five constructs—well-known true, well-known false, obscure true, obscure false and genuinely open—plus ten arithmetic anchors. Every generation is released.

Construct	mean	SD between items	SD / mean
Ethical conflict	0.222	0.211	0.95
Logical paradox	0.117	0.193	1.66
Vagueness	0.044	0.073	1.65
Future contingency	0.011	0.023	2.11
Epistemic ignorance	0.011	0.035	3.16

Table 2: Ten items per construct. The standard deviation between item means equals or exceeds the construct mean in four of the five cases, and is of the same order in the fifth.

3 Item sampling: the rate is a property of the items

That items differ in how much they tell you is the premise of the item-response work applied to NLP leaderboards and test sets [Rodriguez et al., 2021, Vania et al., 2021]; what we add is the consequence for a rate reported over a handful of them. Figure 1 shows the rate for every item separately. The pattern is the finding: *the spread between items of the same construct is as large as the effect.*

The consequence is quantitative. A rate estimated from k items has a standard error of approximately $s/\sqrt{k}$, where s is the between-item standard deviation of Table 2. Taking the largest construct, $s = 0.211$:

At one item the 95% interval half-width is ± 0.41. The interval is wider than most effects anyone reports, which is another way of saying that a single-item probe of a construct like this one is, to a first approximation, uninformative about the construct. Ten items give ± 0.13. Reaching ± 0.05 takes 69.

The worked case. The corpus we borrow was preceded by a study using one statement per construct, which reported a rate of 0.661 for the construct we call ethical conflict [Smarandache and Leyva-Vázquez, 2026b]. Measured over ten items the same quantity is 0.222, and the per-item values run from 0.000 to 0.667. The published figure is not a mismeasurement of its sentence; it is an accurate

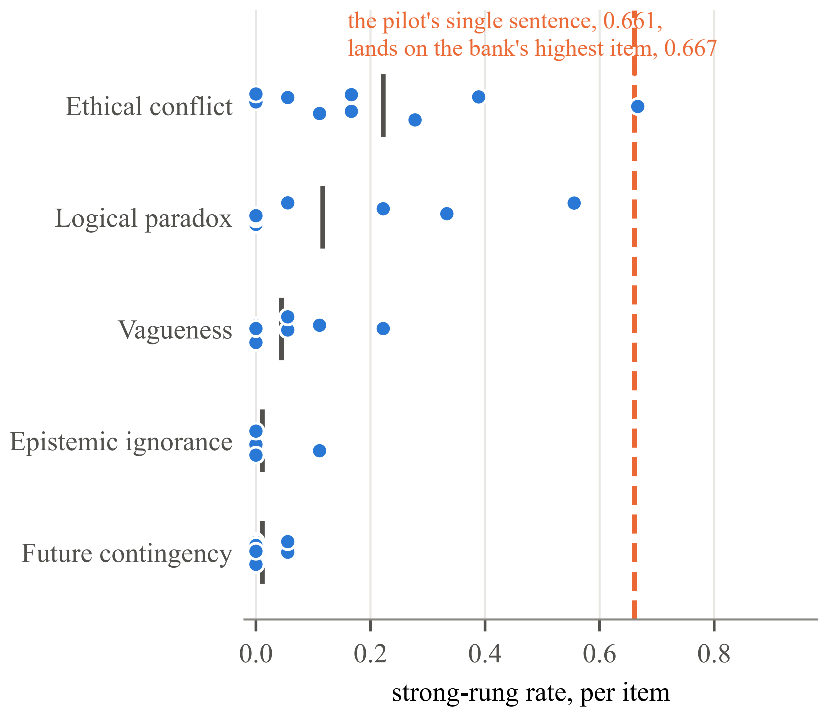

Figure 1: Rate for each of the ten items in each construct; the short bar is the construct mean. Where the effect is large the spread is large. The dashed line is a published figure obtained from a single statement, which lands on the highest of the ten items measuring the same construct.

measurement of the most extreme of ten, reported as though it described the construct. This is the failure mode in its purest form, and we suspect it is common precisely because nothing in the single-item result looks wrong.

Design rule. Report the between-item standard deviation, not only the mean. If it is of the order of the mean—and in five of five constructs here it was—then a difference between two constructs is not interpretable without an interval that resamples *items*, not evaluations. In our corpus the aggregate contrast survives such a bootstrap ($\Delta = 0.178$, $[0.058, 0.312]$) while the contrast against the nearest single construct does not ($\Delta = 0.107$, $[-0.060, 0.272]$), and only the item-clustered interval reveals the difference between those two situations.

4 Instruction effects: ablate the framing, not the sentence

Elicited evaluations carry an instruction that explains what is being asked. The instruction is part of the measurement, and when the behaviour of interest is one the instruction mentions, the natural worry is that the instruction produces it.

In our corpus the worry is concrete. The system message reads, in part: *"These dimensions are NOT constrained to sum to 1.0. A statement can be simultaneously partially true, partially false, partially indeterminate and partially neutral."* The study then measures how often models report values whose

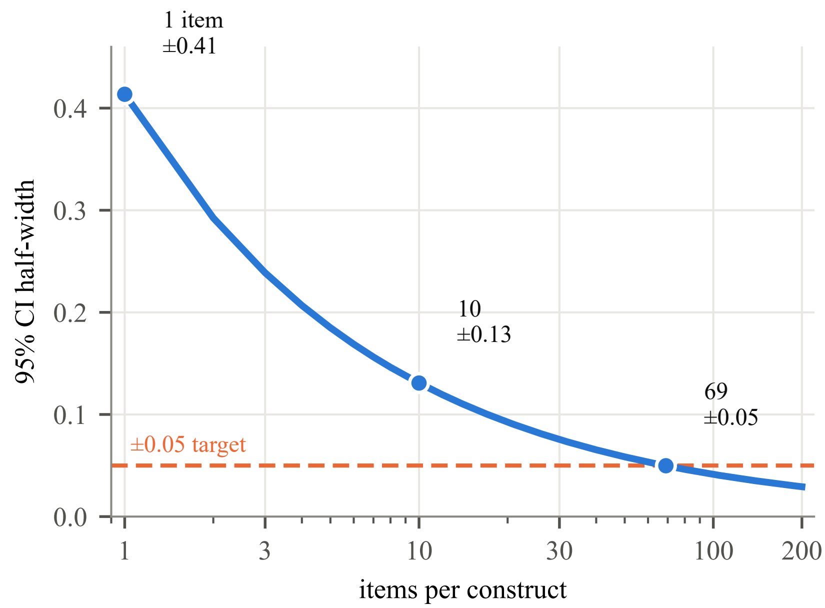

Figure 2: Precision of a construct-level rate as a function of items, at the between-item standard deviation we measure. One item gives a 95% interval half-width of ± 0.41 on a scale that runs from 0 to 1.

sum exceeds one. The instruction names the outcome.

The natural response is to delete the offending sentences and re-run. We did that, and we also ran a second condition that removes the entire framing—the expert role, the explanation of what each dimension means—leaving a generic instruction to answer in JSON. Same 60 items, same six models, same user message, 360 elicitations each.

Deleting the permission does almost nothing. The rate moves from 0.077 to 0.060 with intervals that overlap across most of their length; on the largest construct, from 0.200 to 0.155. Models exceed the bound at nearly the same rate when nothing has told them they may. Had we stopped there, we would have concluded that the instruction was innocent.

Removing the framing takes the rate to 0.004 and raises exact normalization from 2.7% to 30.4%. But this condition does not withdraw a permission; it withdraws the question. A model asked for four unexplained numbers treats them as a distribution, because nothing has told it they are anything else. The honest conclusion is neither that the instruction manufactures the result nor that the result is independent of how it is asked: the behaviour is a property of the framed question, and does not survive being asked in another language.

Why one ablation is not enough. The two conditions disagree, and either alone would have misled us. The minimal ablation says the instruction is innocent; the maximal one says it is everything. Only the pair locates the effect—in the framing, not the licence—and only the pair makes clear that the maximal condition is not a control but a different experiment.

Design rule. Run two nested ablations, not one. Remove the smallest span of text that names the outcome; separately, remove the framing that makes the question intelligible. Report both. If they disagree, the effect lives between them, and the paper's claim must be scoped to the framed question rather than

	full instruction	permission deleted	framing removed
Target rate, contested items	0.077	0.060	**0.004**
95% CI	[.052,.113]	[.038,.093]	[.001,.020]
Target rate, largest construct	0.200	0.155	**0.000**
Sum exceeds one	0.973	0.947	0.696
Sum equals exactly one	0.027	0.049	**0.304**
Rate on anchors	0.000	0.000	0.000
Unparseable responses	0.6%	6.4%	7.5%

Table 3: Two nested ablations of the instruction. Deleting the sentences that name the outcome changes little. Removing the framing changes everything—and also withdraws the question.

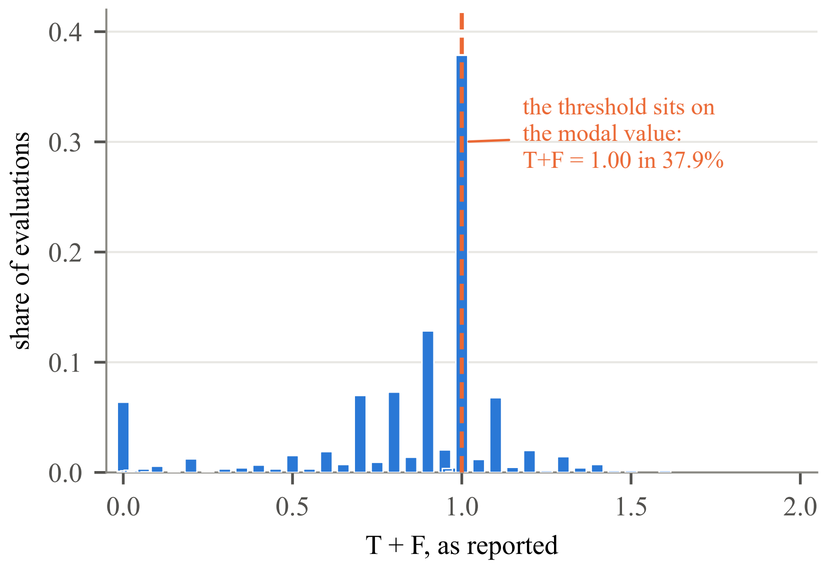

Figure 3: The quantity our decision rule thresholds, as reported. Models answer on a coarse grid, and the threshold falls on the modal value.

to the model. Report parse failures per condition too: ours rose more than tenfold, all of it from one model, which is a fact about that model and not about the manipulation, but a reader cannot know that unless it is broken out.

5 Threshold artifacts: the rule sits where the answers pile up

Language models do not report continuous numbers. Across $7{,}852$ elicited components in this corpus there are 46 distinct values; 97.7% are multiples of 0.05 and 89.5% are multiples of 0.1. The five most common values of one component account for 64% of its mass. This is a property of how models write numbers, not of the quantity being reported, and it is stable across all four conditions we ran.

A decision rule that compares such a quantity against a fixed constant therefore adjudicates a large share of cases by rounding. In our corpus the rule is a strict inequality against 1, and the quantity equals exactly 1.00 in 37.9% of evaluations—the modal value. Figure 3 shows the pile-up sitting on the boundary.

That cutting a continuous quantity into two classes discards information and can mislead is a long-standing result in psychological methods [MacCallum et al., 2002]; what is specific here is that the cut point coincides with the value models return most often. The consequence is not subtle. Replacing the strict comparison with a non-strict one—a change no larger than the resolution at which models answer—moves the rate on the anchor condition, which is the study's control, from 0.000 to 0.778. A result that had read as "the signature never fires on settled material" turns out to be a statement about an inequality sign. The ordering across constructs survives the perturbation, attenuated; the dramatic form of it does not.

Design rule. Before fixing a threshold, plot the distribution of the quantity it cuts. If the cut point coincides with a modal value, report the rate under at least two tie-breaking conventions and treat any conclusion that changes between them as a property of the convention. Where the choice is free, place the threshold off-grid—at 1.025 rather than 1—or use a margin, or report the underlying quantity instead of the label.

6 A fourth, briefly: agreement across model families

One further quantity is worth separating because it is routinely misread [Cohen, 1960, Shrout and Fleiss, 1979]. Studies that use several models as raters report an agreement coefficient over their labels. With one generation per cell at a non-zero temperature, such a coefficient cannot distinguish models disagreeing from a single model answering differently on a rerun.

Our corpus contains a design that separates them, because the pilot has ten repetitions. Over the five contested statements, two repetitions of the same item by the same model under the same wording land on different labels 17.5% of the time ($n =$ 4,032 pairs, 95% CI [9.3, 25.1]); two different models answering the same item, wording and repetition disagree 32.8% of the time ($n =$ 2,240 pairs, 95% CI [13.3, 52.5]). Roughly half of the disagreement attributed to raters is stochastic variation a single rater produces on its own, and that ratio is the stable part: it is 0.53 over the contested statements and 0.52 with the three tautological controls added back, although adding them halves both rates. The intervals are wide because the pilot has five contested statements, which is the first result of this paper applied to itself; the decomposition is reproduced by `family1/analyze_pilot_agreement.py`. The pooled coefficient over the bank, $\kappa = 0.184$ [Fleiss, 1971], should be read as a bound on reproducibility rather than as a measure of inter-model disagreement.

Design rule. Either include repetitions and decompose, or report the coefficient as a reproducibility bound and say why. Do not describe distinct model families as interchangeable raters; they are different instruments, and the coefficient is not inter-rater reliability in the sense the term usually carries.

7 Replication on a second construct family

The three results above come from one corpus about one subject matter. If they are properties of the elicited design rather than of that subject matter, they should reappear in a task that shares the design and nothing else.

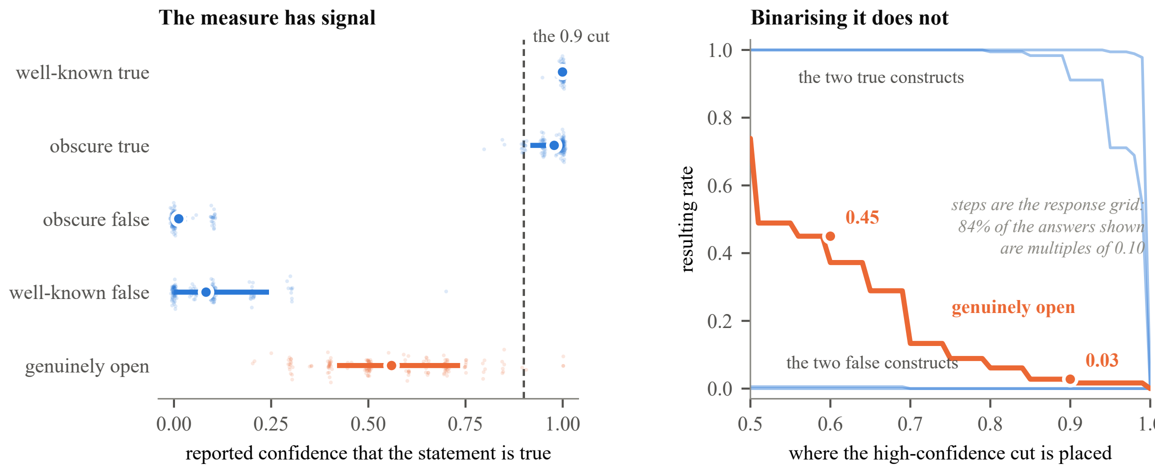

Figure 4: Second bank, the 900 construct elicitations; the 180 anchor elicitations are omitted because they carry no dispersion to show. Left: reported confidence per elicitation, with the item means spanned by the bar. The open-question construct occupies the middle of the scale and spreads across items. Right: the rate that survives binarisation, as a function of where the cut is placed. Moving the cut from 0.60 to 0.90 takes the open-question rate from 0.450 to 0.028 without touching the data.

So we built a second bank. The construct is factual confidence: the model is shown a statement and reports a single number in $[0, 1]$ for how confident it is that the statement is true. This is the most ordinary elicited evaluation there is, and the quantity that the calibration literature reports constantly. Five constructs of ten matched items each—well-known true, well-known false, obscure true, obscure false, and genuinely open questions with no known answer—plus ten arithmetic anchors. Same six model families, same three wordings, same two nested instruction ablations. Nothing about four-valued logic appears anywhere in it.

The bank behaves. Parse failure is 0.0% over $1{,}080$ elicitations, the arithmetic anchors return exactly 1.000 and 0.000, and where ground truth exists the models assign high confidence to true items and low confidence to false ones (Brier error 0.005; we do not estimate a calibration curve, so this is accuracy rather than calibration in the reliability sense). Whatever the bank measures, it is not noise.

The outcome we chose first was the artifact. We initially tracked a *high-confidence rate*: the fraction of elicitations at or above 0.9. That is the domestic analogue of an overconfidence rate and the form in which such results are almost always published. On this bank it is close to useless. Four of the five constructs pin to 0.000 or 1.000 with zero between-item variance, and the one construct with real dispersion—the open questions, where models spread from 0.42 to 0.74 across items and disagree with each other by 0.110—collapses to a rate of 0.028. The signal is not absent. The threshold removes it.

That is Section 5 arriving unbidden in a new domain, and it is worth stating in the form a practitioner will meet it: *the binarisation is not a summary of the measurement, it is a second measurement, and on this bank it is the one that fails*. We therefore report the continuous quantity as primary throughout this section and keep the rate only to show what it costs.

7.1 Item sampling replicates, and gives the rule its real shape

Measured on the continuous scale, the between-item standard deviation within a construct ranges from 0.001 to 0.112. Converted to a sample size for a ± 0.05 half-width, that is between 1 and 20 items *within a single bank*—and the same calculation on the first corpus returned 69.

This is a stronger form of the rule than Section 3 could state on its own. The finding is not that between-item variance is large. It is that the number of items a construct needs varies by a factor of nearly seventy across ordinary evaluation targets, with no way to tell in advance which kind you have. Our own open-question construct needed 20 items and we gave it 10; its interval is correspondingly wide, $[0.496,\ 0.624]$.

7.2 Instruction effects do not replicate in size, and the reason sharpens the rule

The second bank carries the same two nested ablations. Condition A$'$ deletes the sentence that licenses hedging—*you are NOT required to commit to a verdict: when the evidence is insufficient or the question is genuinely open, report a confidence near 0.5*. Condition B$'$ deletes the surrounding role and scale as well, leaving only *you are a careful evaluator*. Both are paired: same items, same models, same wording, 360 elicitations each, 100% parse rate in all three conditions.

The first half replicates exactly. Deleting the licence moves the open-question construct from 0.551 to 0.531, a paired difference of -0.020 with a 95% interval of $[-0.052,\ +0.009]$. That is the first corpus's result again: the sentence that names the measured behaviour is worth almost nothing.

The second half does not replicate in the mean. Deleting the entire framing moves the same construct to 0.533—a paired difference of -0.018, interval $[-0.044,\ +0.009]$. On the first corpus the corresponding ablation took the headline quantity from 0.077 to 0.004. Here it does nothing.

Why, and why it was predictable. The two system messages are not equally load-bearing, and the asymmetry is structural rather than empirical. In the first corpus the system message carried the whole conceptual apparatus—four independent components not constrained to sum to one—which the model cannot reconstruct from the question alone; removing it removed the task. Here it carries a role label and a scale that the user turn already restates, since the question is *how confident are you that this statement is true* and already asks for *a single number in* $[0.0, 1.0]$. There was very little in it to delete.

We state this because it is the useful generalisation, and because stating it after seeing the number would be worth less: *framing is worth what the model could not have inferred without it*. For an exotic elicited construct that is everything. For a familiar one it is close to nothing. The design rule is unchanged—run both nested ablations—and is now better motivated, because the two corpora are the two regimes and nothing short of running the ablation distinguishes them in advance.

A second reason the mean was the wrong summary. The instruction does change behaviour on this bank. It does not change the mean. On the open-question construct the share of answers at exactly 0.50—the value the licence sentence explicitly names—falls from $17/60$ under the full instruction to $9/60$ under *both* ablations, a paired difference of -0.133 with intervals $[-0.233,\ -0.033]$ for the minimal ablation and $[-0.217,\ -0.067]$ for the full one. The standard deviation of the same answers rises from 0.161 to between 0.195 and 0.219. The instruction is read and obeyed; it redistributes mass symmetrically about the midpoint, so the mean absorbs it and reports nothing.

	continuous		rate at ≥ 0.9	
Construct	mean	SD b/w models	rate	SD b/w models
well-known true	0.999	0.001	1.000	0.000
obscure true	0.978	0.017	0.983	0.031
obscure false	0.012	0.022	0.000	0.000
well-known false	0.083	0.049	0.000	0.000
genuinely open	0.560	0.110	0.028	0.054

Table 4: The same five constructs measured two ways. On the continuous scale the open-question construct is the one with dispersion to explain; after binarisation at 0.9 it is indistinguishable from the two false constructs, all three reading as a rate near zero. The dispersion column is the mean across the ten items of the standard deviation of the six model means, wordings averaged within model.

The resampling unit, declared. All intervals in this section resample *items* with replacement and carry every model's response to a drawn item with it. The item is the sampling unit because six models answer the same statement, so their responses are not independent draws. Resampling rows instead treats them as independent and, on this bank, gives $[-0.267, +0.000]$ and $[-0.217, -0.050]$ for the same two contrasts. We report the item-clustered version and name the alternative because the choice changes whether one interval excludes zero, and a paper about design decisions that produce results should not leave its own undeclared. The comparison is reproduced by `family2/boot_units.py`.

Sixty elicitations per cell and one wording remain a thin base, and we would not build an argument on this contrast alone. We report it because the direction is identical under both ablations and because the methodological point does not depend on it: *an ablation that reads null on the summary statistic can be large on the distribution the statistic summarises*. Section 4 would have concluded that the framing does not matter here. It matters; it moves the shape and not the location.

Design rule. Run both nested ablations, and evaluate each against more than one functional of the response distribution—at minimum a location statistic and one shape statistic such as the mass on a focal value or the dispersion. Report the ablation as null only if it is null on both.

7.3 Threshold artifacts replicate, larger

Models answer this bank on the same coarse grid: 26 distinct values across $1{,}080$ elicitations, 95.7% multiples of 0.05 and 86.4% multiples of 0.1, with 33% of answers at exactly 1.00 and 29% at exactly 0.00. Figure 4 shows what a cut does to that.

The size of the artifact is a factor of 16: the open-question rate is 0.450 at a cut of 0.60, 0.089 at 0.80, and 0.028 at 0.90. None of those numbers is more correct than the others, and a paper reporting any one of them alone would be reporting its own threshold.

7.4 The analogous quantity in the second family, and why it is not the same one

The second bank has no repetitions, so it cannot decompose rater variance the way Section 6 does, and we do not claim it replicates that decomposition. What it has instead is three wordings, which isolates a different component: how much a single model moves when the question is rephrased. That movement is 0.021 in mean absolute confidence, against 0.048 between different models on the same item, a ratio of 0.44. The first corpus gives 0.52 for repetition against models. The two numbers are not the same quantity—one is binary label disagreement across reruns, the other continuous movement across wordings—and we put no weight on their closeness. What the two designs share is only the qualitative conclusion, that a substantial share of what gets attributed to model disagreement is not disagreement between models.

The new observation is about pooling. Between-model dispersion on this bank runs from 0.001 on well-known truths to 0.110 on open questions, a factor of 89. Both figures are the mean, across items, of the standard deviation of the six model means for that item, with the three wordings averaged within model before the spread is taken; computing the same quantity per item-wording cell instead raises the open-question value to 0.120, and we name the unit because the two are not interchangeable. An agreement coefficient computed over a whole bank averages the constructs together and reports a number that describes no construct in it. Agreement is a property of the item, not of the panel.

Design rule. Report agreement per construct, or not at all. If a single pooled coefficient is reported, state the range it was pooled over.

What the second family shows. Three of the four quantities reappear in a task with no shared content, and the one that reappears largest is the one we had ranked third. The fourth, instruction framing, does not reappear at all in the size the first corpus gave it—and that non-replication is the most useful thing in this section, because it identifies what the effect depends on. Framing is worth what the model could not have inferred without it, which is everything for an exotic construct and nearly nothing for a familiar one.

What transfers, then, is the protocol and not a single magnitude. The item count moves from 69 to 20; the threshold effect is the difference between 0.000 and 0.778 on a control condition in the first family and a factor of 16 on the dispersed construct in the second; the framing effect falls from decisive to undetectable in the mean while remaining visible in the shape. Every one of those numbers would have been wrong if borrowed from the other family. That is the argument for measuring them rather than citing them, and it is the reason we ran a second bank instead of asserting that the first one generalises.

8 What this does and does not show

Three magnitudes, two construct families, and no magnitude that transfers between them. Item sampling sets a floor of roughly seventy items for a ± 0.05 estimate in the first family and twenty in the second. Threshold placement is worth the difference between 0.000 and 0.778 on a control condition in the first, and a factor of sixteen in the second. Instruction framing is worth everything in the first and nothing in the mean in the second, where it still moves the shape of the response without moving its location; the reason—framing is worth what the model could not have inferred without it—is the most portable thing in the paper. What generalises is the obligation to measure these three quantities, not any value they took here.

The limits are worth stating plainly. Six models at one point in time, one generation per cell in the main studies, and between-item standard deviations estimated from ten items each: these are offered as an order of magnitude, not as constants. The within-cell variation that matters for the first and third results is estimated from the pilot rather than measured in the bank. The shape finding in Section 7.2 rests on sixty elicitations per cell and is reported as an indication. And nothing here licenses any claim about model internals: every quantity is a property of text produced under an instruction.

The construct labels are ours, and the results do not rest on them. We wrote the items and we assigned them to constructs, so a reader is entitled to ask who says that these ten statements measure ethical conflict rather than something else. That question is the classical one of construct validity [Cronbach and Meehl, 1955, Messick, 1995], and it arrives here in the form measurement theory gives it for computational systems [Jacobs and Wallach, 2021]. The honest answer is that we do, and that the assignment has not been validated against independent judges. What we can say is more limited than we first wrote, and the limit runs in a direction worth setting out.

In the first family the second and third results are manipulations of the instrument—an instruction deleted, a threshold moved—applied to a fixed set of items, and there the labels are not load-bearing: the same statements are asked twice and the difference is the effect.

In the second family they are load-bearing, and we were wrong to imply otherwise. Both effects live entirely on the open-question construct. On the four settled constructs the answers are pinned near 0 or 1, where moving a threshold changes nothing and deleting an instruction changes nothing, and Table 4 shows exactly that. So the existence of those two effects depends on that subset really consisting of questions with no settled answer. If those ten items are not genuinely open, the effects do not shrink—they have nowhere to occur. That is a construct-validity dependency at the centre of the replication, and the classification instrument we release covers the first family's items and not these, which is the gap a reader should hold against us first.

The first needs a qualification we would rather state than have extracted from us. A group wrongly named still has the dispersion we report, so the measurement stands; but the rule drawn from it—that reaching a ± 0.05 half-width takes sixty-nine items—is an inference about estimating a *construct-level* quantity, and it presupposes that the items sample one. If they do not, the between-item standard deviation mixes variability among items that do belong to a construct with heterogeneity among items that do not, and we cannot say in which direction that moves the estimate: a heterogeneous group is not guaranteed to be more variable than the construct it was meant to sample. What follows is weaker and more useful than an upper bound. The sixty-nine is a property of *this bank*, not a constant of the construct, and it stays that way until the assignment is validated. Which is the rule anyway: measure the dispersion in your own bank rather than borrow ours. Construct labelling would matter for a claim of the form "models behave differently on ethical conflict than on vagueness", and the one such contrast we report, in the first family, we report as not established. To let others check the labelling rather than take it on trust, the release includes two blind classification instruments, one per family: the items in randomised order with the anchors mixed in unannounced, the category definitions, and the script that returns agreement between raters and against our assignment. The second family gets a narrower question than the first—whether a statement has a known answer or is genuinely open—because that is the distinction its results rest on. Neither has been run.

What the results do support is narrow and, we think, useful: that these three failure modes are not hypothetical, that their sizes are comparable to the effects such studies report, and that the ablation protocol in Section 4 would have led us to the wrong conclusion if we had run only its first half.

Declarations

Ethics approval and human participants. This study involved no human or animal participants. Every observation in both construct families is a text generation produced by a commercial language model through a programmatic interface, elicited by prompts written by the author. No survey, interview, annotation task or crowdworker judgment was collected, and no personal data were processed. Ethics committee approval and informed consent were therefore not applicable.

Use of language models. Language models are the object of study here rather than an aid to writing: the corpus consists of their outputs, collected between 24 and 26 August 2026 through a single router at temperature 1.0. Vendors and model families are named in Section 2, and every released generation records the model family that produced it. Exact provider version strings are recorded only for the pilot file; the router resolved them at call time and the remaining files carry the family label rather than the resolved version. Collection dates are given here and not per record. These results are properties of specific model versions at a specific date and should not be assumed stable across releases, so this is a limitation of the release rather than a detail: an exact re-run of the same versions is not possible from what we published.

Competing interests. The Family I corpus was collected for two companion studies co-authored by the present author, and one published figure re-analysed here (0.661) comes from that work. This paper reports that the figure does not survive a larger item sample. No other competing interests are declared.

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Data and code availability

Both item banks, the elicitation and analysis scripts, and all 7,920 raw generations including response text are public at `https://github.com/mleyvaz/elicited-evaluation-design`. Every figure and table here is produced by released code from released data. Code is MIT-licensed; the banks and the generations are CC BY 4.0.

The Family I generations were collected for the companion studies rather than for this paper. They are reproduced in full so that every number here can be checked in one place, with their provenance stated in the repository; the citable source for that bank, together with an independent re-analysis script written by a party that did not author the studies, remains `https://github.com/mleyvaz/paraconsistent-signature-itembank`.

The second bank is released in the form in which it was analysed, including the first version of its analysis script. That script took the high-confidence rate as its primary outcome, which Section 7 reports as the artifact it turned out to be; we keep it in the repository rather than quietly replacing it, because the sequence is part of the evidence for the third design rule.

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References

Samuel R. Bowman and George E. Dahl. What will it take to fix benchmarking in natural language understanding? In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, pages 4843–4855, 2021. URL `https://aclanthology.org/2021.naacl-main.385/`.

Jacob Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20 (1):37–46, 1960. doi: 10.1177/001316446002000104.

Lee J. Cronbach and Paul E. Meehl. Construct validity in psychological tests. *Psychological Bulletin*, 52(4): 281–302, 1955. doi: 10.1037/h0040957.

Lee J. Cronbach, Goldine C. Gleser, Harinder Nanda, and Nageswari Rajaratnam. *The Dependability of Behavioral Measurements: Theory of Generalizability for Scores and Profiles*. Wiley, New York, 1972.

Joseph L. Fleiss. Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382, 1971. doi: 10.1037/h0031619.

Abigail Z. Jacobs and Hanna Wallach. Measurement and fairness. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 375–385, 2021. doi: 10.1145/3442188.3445901.

Hankyeol Kim and Pilsung Kang. Same answer, different confidence: Protocol sensitivity in LLM confidence calibration. *arXiv preprint arXiv:2605.27752*, 2026. URL `https://arxiv.org/abs/2605.27752`.

Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022. URL `https://openreview.net/forum?id=8s8K2UZGTZ`.

Robert C. MacCallum, Shaobo Zhang, Kristopher J. Preacher, and Derek D. Rucker. On the practice of dichotomization of quantitative variables. *Psychological Methods*, 7(1):19–40, 2002. doi: 10.1037/1082-989X.7.1.19.

Samuel Messick. Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist*, 50(9):741–749, 1995. doi: 10.1037/0003-066X.50.9.741.

Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. *arXiv preprint arXiv:2411.00640*, 2024. URL `https://arxiv.org/abs/2411.00640`.

Pedro Rodriguez, Joe Barrow, Alexander Miserlis Hoyle, John P. Lalor, Robin Jia, and Jordan Boyd-Graber. Evaluation examples are not equally informative: How should that change NLP leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL-IJCNLP)*, pages 4486–4503, 2021. URL `https://aclanthology.org/2021.acl-long.346/`.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

Patrick E. Shrout and Joseph L. Fleiss. Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2):420–428, 1979. doi: 10.1037/0033-2909.86.2.420.

Florentin Smarandache and Maikel Y. Leyva-Vázquez. One item is not a phenomenon: Separating ethical content from syntactic marking in paraconsistent signature detection, 2026a. Submitted to the AAAI 2026 Fall Symposium Series. Item bank and generations released under CC BY 4.0.

Florentin Smarandache and Maikel Y. Leyva-Vázquez. Is the ladder measurable? graded paraconsistency tested on eight statements and then on a bank, 2026b. Submitted to the AAAI 2026 Fall Symposium Series.

Clara Vania, Phu Mon Htut, William Huang, Dhara Mungra, Richard Yuanzhe Pang, Jason Phang, Haokun Liu, Kyunghyun Cho, and Samuel R. Bowman. Comparing test sets with item response theory. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL-IJCNLP)*, pages 1141–1158, 2021. URL `https://aclanthology.org/2021.acl-long.92/`.

Abel Yagubyan. The coin flip judge? reliability and bias in LLM-as-a-judge evaluation. *arXiv preprint arXiv:2606.13685*, 2026. URL https://arxiv.org/abs/2606.13685.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*, 2023.